MASTERS PROJECT MILESTONE REPORT

Name of the student:

Name of the supervisor(s):

Title of the project (may be tentative): Poisson Surrogate for High-Dimensional Multinomial Regression Models

Summary: This article is motivated by Taddy, M (2015) and Lee et al (2017) which have demonstrated that the multinomial regression models are used for datasets including counts in a massive number of categories. Dependence among responses can be incorporated through group-level random effects with grouped observations, leading to multinomial mixed models. However, Lee, Green & Ryan (2017) showed that closed-form likelihoods for multinomial mixed models do not exist regardless of the random effect distribution, except for the special case with no covariates. This lack of tractable likelihood representation leads to the computational obstacle for fitting multinomial mixed models.

"Poisson trick" (Lee et al., 2017) proposed a surrogate model for reformulating multinomial mixed models. Let $Y_i = (Y_{i1}, \ldots, Y_{iK})$ denotes a multinomial count vector with the fixed sum $m_i = \sum_{k=1}^{K} Y_{ik}$. Suppose that the multinomial sums to be random according to a Poisson distribution, the likelihood of multinomial is identical to the likelihood of surrogate Poisson model up to an additive constant. Hence the maximum likelihood estimates, asymptotic variances and tests for the multinomial can be recovered under Poisson surrogate model (Richards, 1961).

Previous studies have focused on low-dimensional datasets, where the number of covariates p is smaller than the sample size n. In contemporary applications, however, one encounters regimes in which $p \gg n$ or where the total parameter dimension grows with both the number of predictors and response categories. In such settings, classical multinomial methodology faces serious difficulties including non-identifiability, instability of maximum likelihood estimation, overfitting and increased computational cost.

Therefore, the "Poisson trick" should be viewed as a useful start for further methodological development, but not as a complete solution to high-dimensional modelling. Structuring the multinomial likelihood makes it possible to incorporate regularity constraints on the parameter matrix, which are essential when the effective model complexity is too large for unregularized estimation.

Motivated by this perspective, this article will investigate high-dimensional extensions of multinomial modelling based on Poisson surrogate framework. The main goal is to find a tractable likelihood reformulations that can be combined with suitable regularization to achieve statistically reliable and computationally scalable estimation. We aim to retain the computational advantages of the Poisson reformulation while addressing the statistical

challenges inherent in modern multinomial regression with many covariates and potentially complex dependence structures.

Student's signature **Supervisor's signature**

Date: